must be submitted to the committee on or before the end of the first week of classes of the first semester.

The **Master of Engineering (non-thesis)** requires the following courses in addition to the Core Competencies:

GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGN599	INDEPENDENT STUDY	6.0
Candidates must also take at least three of the following:		
GEGN563	APPLIED NUMERICAL MODELLING FOR GEOMECHANICS	3.0
GEGN570	CASE HISTORIES IN GEOLOGICAL ENGINEERING AND HYDROGEOLOGY	3.0
or GEGN673	ADVANCED GEOLOGICAL ENGINEERING DESIGN	
GEGN573	GEOLOGICAL ENGINEERING SITE INVESTIGATION	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
or GEGN580	APPLIED REMOTE SENSING FOR GEOENGINEERING AND GEOSCIENCES	
or GEGN568	POINT CLOUD DATA ANALYSIS IN EARTH SCIENCE AND ENGINEERING	
GEGN671	LANDSLIDES: INVESTIGATION, ANALYSIS & MITIGATION	3.0

Electives and course substitutions are approved by the advisor. Possibilities for other electives include graduate- level rock mechanics and rock engineering, soil mechanics and foundations, groundwater, site characterization, geographical information systems (GIS), project management and geophysics, for example.

Program Details for Graduate Certificates

Certificate and Degree Requirements

We offer two graduate certificates and a professional master's degree (non-thesis). The courses taken for certificate degrees can be used toward the professional master's degree.

The Graduate Certificate Programs in Economic Geology and Exploration Methods outlined below may be completed by individuals already holding undergraduate or advanced degree in geology or a related field and have at least two to three years of professional experience. The programs are comprised of:

Course Work	12.0 Credits
Total Semester Hrs	12.0 Credits

Graduate Certificate of Economic Geology

Students working toward a Graduate Certificate of Economic Geology are required to take at least 6 credits out of the following courses; courses cannot be used in fulfilling the requirements of other certificates:

Core courses:

GEOL513	HYDROTHERMAL GEOCHEMISTRY	3.0
GEOL521	FIELD AND ORE DEPOSIT GEOLOGY	3.0
GEOL524	ORE DEPOSIT MODELS AND EXPLORATION STRATEGIES	3.0

Students working toward a Graduate Certificate of Economic Geology can choose up to 6 credits out of the following courses' courses cannot be used in fulfilling the requirements of other certificates:

Electives:

GEOL505	ADVANCED STRUCTURAL GEOLOGY	3.0
GEOL520	NEW DEVELOPMENTS IN THE GEOLOGY AND EXPLORATION OF ORE DEPOSITS	2.0
GEOL523	REFLECTED LIGHT AND ELECTRON MICROSCOPY	2.0
GEOL628	ADVANCED IGNEOUS PETROLOGY	3.0
GEOL645	VOLCANOLOGY	3.0
CHGC503	INTRODUCTION TO GEOCHEMISTRY	3.0

Graduate Certificate of Exploration Methods

Students working toward a Graduate Certificate of Economic Geology are required to take at least 6 credits out of the following courses; courses cannot be used in fulfilling the requirements of other certificates:

Core courses:

GEGX571	GEOCHEMICAL EXPLORATION	3.0
GEOL520	NEW DEVELOPMENTS IN THE GEOLOGY AND EXPLORATION OF ORE DEPOSITS	2.0
GEOL523	REFLECTED LIGHT AND ELECTRON MICROSCOPY	2.0
GPGN598	GEOPHYSICS FOR MINERAL EXPLORATION	3.0

Students working toward a Graduate Certificate of Economic Geology can choose up to 6 credits out of the following courses; courses cannot be used in fulfilling the requirements of other certificates:

Electives:

GEGN501	MINERAL DEPOSITS	
GEGN502	MINERAL EXPLORATION DESIGN	
GEGN532	GEOLOGICAL DATA ANALYSIS	3.0
GEGN575	APPLICATIONS OF GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEGN588	ADVANCED PLANETARY GEOGRAPHIC INFORMATION SYSTEMS	3.0
GEOL501	APPLIED STRATIGRAPHY	4.0
GEOL514	BUSINESS OF ECONOMIC GEOLOGY	3.0
GEOL521	FIELD AND ORE DEPOSIT GEOLOGY	3.0
GEOL524	ORE DEPOSIT MODELS AND EXPLORATION STRATEGIES	3.0
GEOL526	PLATE TECTONICS	3.0
GEOL555	STRUCTURAL FIELD RESEARCH	4.0
GEOL598	SEMINAR IN GEOLOGY OR GEOLOGICAL ENGINEERING (Skarns and Related Deposits)	3.0
GEOL601	CORE TO OUTCROP STRATIGRAPHY	2.0
MNGN510	FUNDAMENTALS OF MINING AND MINERAL RESOURCE DEVELOPMENT	3.0
MNGN528	MINING GEOLOGY	3.0
CHGC504	METHODS IN GEOCHEMISTRY	3.0